A Formal-Development Proof of Prescribed-Edge Induced-Matching Packing

Route B: a mathematical exposition of the original StrongEdge proof

10 October 2026

Abstract. Let G be a finite simple graph of maximum degree at most four. We prove that every prescribed set $F \subseteq E(G)$ contains an induced matching M of the original graph such that $|F| \leq 20|M|$. Equivalently, every induced subgraph of the square of the line graph of G has an independent set containing at least one twentieth of its vertices. The proof establishes five clique bounds, according to the degrees of their vertices in the full conflict graph, and combines them with a private-neighbourhood counting argument. All structural bounds needed for the argument are proved below. The constant is sharp by the familiar doubled five-cycle. The full-edge-set bound and this extremal example are already present in Joos and Nguyen (2016); the prescribed-set formulation retains all conflicts of the original graph. The proof follows the original StrongEdge dependency structure: one-sided covers, specialized four-cycle budgets, branch estimates and a five-field packing criterion. No claim of priority over the entire literature is made.

Keywords. Induced matching; prescribed edge set; square of a line graph; Hall ratio; maximum degree four; formal verification.

1. Statement and context

This manuscript gives the full mathematical argument corresponding to the unchanged original **StrongEdge** proof. It is self-contained: no result from the companion Route A manuscript is assumed. The two expositions share the same overall clique-bound and private-neighbourhood strategy. Here the intermediate statements and case interfaces follow the original formal implementation, including bounds that the code deliberately states more weakly than its internal estimates permit.

All graphs in this manuscript are finite, simple and undirected. An **induced matching** in G is a set of pairwise vertex-disjoint edges such that no edge of G joins endpoints of two different chosen edges. In particular, an edge outside a prescribed candidate set is still allowed to prohibit a pair of candidate edges.

Theorem 1.1 (Prescribed-edge packing). Let $\Delta(G) \leq 4$ and let $F \subseteq E(G)$. There is an induced matching M in G satisfying

$$M \subseteq F, \qquad |F| \leq 20|M|. \tag{1.1}$$

Thus $|M| \geq \lceil |F|/20 \rceil$. The constant 20 cannot be replaced by any smaller real constant.

For distinct edges $e, f \in E(G)$, say that e and f **conflict** if they share an endpoint or if an endpoint of e is adjacent in G to an endpoint of f. The **conflict graph** H_G has vertex set $E(G)$ and this adjacency relation; it is the square $L(G)^2$ of the line graph. Independent sets of H_G are exactly induced matchings of G. Write $c(e) = d_{H_G}(e)$ for the full conflict degree of an edge.

If $\alpha(J)$ is the independence number of a graph J, its Hall ratio is

$$\rho(J) = \max_{\emptyset \neq X \subseteq V(J)} \frac{|X|}{\alpha(J[X])}. \tag{1.2}$$

We set $\rho(J) = 0$ when J is empty. Theorem 1.1 is equivalent to $\rho(H_G) \leq 20$. Here $H_G[F]$ is obtained by restricting the vertices of the conflict graph, while its edges retain their meaning in the full original graph.

Joos and Nguyen [1, Theorem 1.1 and the corollary on p. 156] prove

$$\nu_s(G) \geq \frac{n(G) - i(G) - c_5(G)}{9}, \tag{1.3}$$

where $i(G)$ counts isolated vertices and $c_5(G)$ counts components isomorphic to the doubled five-cycle. They explicitly deduce $\nu_s(G) \geq |E(G)|/20$, and obtain $|E(G)|/18$ when there is no such component. Their sharpness example and the optimality of the full-edge-set constant are therefore prior results. Theorem 1.1 extends the full-edge-set 20-bound to prescribed sets; it does not subsume all the more refined assertions in (1.3).

The distinction between these formulations is visible on the path $0-1-2-3$. For $F = \{01, 23\}$ the two candidates conflict through 12, even though $12 \notin F$. Deleting 12 changes the question. Consequently, applying a full-edge-set theorem to the graph with edge set F is not a proof of Theorem 1.1. Likewise, a large induced matching chosen without regard to F may avoid F entirely.

The private-neighbourhood and exchange ideas used here have methodological precedents; see, for example, the local-search analysis of Fürst, Leichter and Rautenbach [3]. The strong edge-colouring bound 21 of Huang, Santana and Yu [2] directly gives the analogous prescribed-set bound with 21. The argument below proves the bound 20 by clique estimates and counting. It does not construct a strong edge-colouring with 20 colours or prove a fractional strong chromatic bound of 20. In general a Hall-ratio bound cannot be promoted to the corresponding fractional chromatic bound [4].

2. Edge covers and two elementary clique bounds

A **conflict clique** is a set $Q \subseteq E(G)$ of pairwise conflicting edges. Throughout the structural part of the proof, Q denotes such a clique and

$$q = |Q|, \qquad d_Q(x) = |\{e \in Q : x \in e\}|.$$

The number $d_Q(x)$ is the degree in the selected edge set, whereas $d_G(x)$ counts all original edges. These quantities will not be interchanged. For $S \subseteq V(G)$, let $e_G(S)$ be the number of original edges with both endpoints in S.

Lemma 2.1 (Endpoint-neighbour cover). For $uv \in E(G)$ put

$$C(uv) = (N_G(u) \setminus \{v\}) \cup (N_G(v) \setminus \{u\}).$$

Then $u, v \notin C(uv)$, $|C(uv)| \leq 6$, and

$$c(uv) + e_G(C(uv)) \leq \sum_{x \in C(uv)} d_G(x) \leq 24. \tag{2.1}$$

For $uv \in Q$ one also has

$$q \leq 1 + \sum_{x \in C(uv)} d_Q(x). \tag{2.2}$$

Proof. Every edge conflicting with uv meets $C(uv)$. If it shares an endpoint with uv, its other endpoint lies in the cover. Otherwise a length-two path in the line graph supplies an original edge joining an endpoint of uv to an endpoint of the other edge, and that latter endpoint lies in the cover. The converse implication is not needed here.

For any selected edge set D and vertex set S, let $m_D(S)$ count edges of D meeting S, and let $i_D(S)$ count those with both endpoints in S. Counting incidences gives the exact identity

$$\sum_{x \in S} d_D(x) = m_D(S) + i_D(S).$$

The conflict neighbours of uv are a subset of the original edges meeting $C(uv)$. Applying this identity with $D = E(G)$ proves the first inequality of (2.1). Each erased neighbour set has at most three vertices, giving the upper bound twenty-four. With $D = Q$, the inclusion of $Q \setminus \{uv\}$ in the edges meeting the cover gives (2.2). This proof only uses inclusions between these edge families. ■

Lemma 2.2 (Triangles and local slack). All edges have $c(e) \le 24$. An edge in a triangle has $c(e) \le 20$. If $c(uv) \ge t$, then $e_G(C(uv)) \le 24 - t$. If a triangle edge has $c(uv) \ge 20$, its cover has no internal edge. If $c(uv) \ge 21$, then $d_G(u) = d_G(v) = 4$ and uv is in no triangle.

Proof. A common neighbour of u, v reduces the cover size to at most five, so (2.1) gives the triangle assertion and the empty-cover assertion at equality. If either endpoint has degree at most three, the cover likewise has size at most five. The other statements follow from (2.1). ■

Lemma 2.3 (Selected support degree at most three). If $d_Q(x) \le 2$ for every x, then $q \le 13$. If $d_Q(x) \le 3$ for every x, then $q \le 18$.

Proof. The empty clique is immediate. Otherwise (2.2) gives $q \le 1 + 6s$ when all selected degrees are at most s. It remains to exclude $q = 19$ for $s = 3$.

Suppose $q = 19$. For every selected edge uv, equality throughout (2.2) forces $|C(uv)| = 6$ and $d_Q(a) = 3$ for every $a \in C(uv)$. We show that every vertex supporting a selected edge has selected degree three. Let $xy \in Q$. Since its cover has size six, there is a neighbour $a \ne y$ of x. This vertex lies in $C(xy)$ and hence has three selected incident edges. Choose a selected edge ab with $b \ne x$. Now $x \in C(ab)$, so $d_Q(x) = 3$ as well. Thus every selected degree is either zero or three. The handshake identity $\sum_x d_Q(x) = 2q = 38$ would make 38 divisible by three, a contradiction. ■

The next lemma will be used inside a set of vertices while retaining every original edge between those vertices.

Lemma 2.4 (A degree-two restriction). Suppose $P \subseteq V(G)$, every edge of a conflict clique Q' has both endpoints in P, and $|N_G(x) \cap P| \le 2$ for every $x \in P$. Then $|Q'| \le 5$. If there is a set $D \subseteq P$ such that each selected edge in Q' has exactly one endpoint in D, then $|Q'| \le 4$. Original edges outside Q' need not respect this bipartition.

Proof. For a selected edge uv, use the restricted cover $C(uv) \cap P$. Since both endpoints of every selected edge lie in P, every conflicting selected edge other than uv meets this restricted cover. Each endpoint has at most two original neighbours in P, so the restricted cover has at most two vertices. Each of these has selected degree at most two. The incidence covering inequality therefore gives $|Q'| \le 1 + 2 \cdot 2 = 5$.

If equality holds, every cover of a selected edge has two vertices of selected degree two. For a supported vertex x choose $xy \in Q'$. Equality forces x to have a neighbour $a \ne y$ in P; this neighbour lies in the cover and has two selected edges. Choose $ab \in Q'$ with $b \ne x$. Since x belongs to the cover of ab, it too has selected degree two. Thus every supported vertex has selected degree two. If each selected edge crosses D, then $|Q'| = \sum_{x \in D} d_{Q'}(x)$ is even, contradicting $|Q'| = 5$. ■

3. Root notation and counting tools

Choose a vertex r, called the root, and write

$$A = N_G(r), \qquad R = \{a \in A : ra \in Q\}, \qquad s = |R| = d_Q(r).$$

When A is independent, every selected edge has exactly one of the following forms: a root edge ra with $a \in R$; an edge ax with $a \in A$ and $x \notin A \cup \{r\}$; or an edge with both endpoints outside $A \cup \{r\}$. Denote

the second set by K, the third by O, and put $k = |K|$, $o = |O|$. The implementation uses the resulting covering estimate

$$q \le s + k + o. \tag{3.1}$$

Edges in O will be called outer edges. Even without independence of A, every selected edge either meets A or is outer, so

$$q \le \sum_{a \in A} d_Q(a) + o. \tag{3.2}$$

For $x \ne r$ define its **type**

$$T(x) = N_G(x) \cap A.$$

A vertex of type size one is a singleton, of type size two a double, and of type size three a triple. Let $P = \{x \ne r : |T(x)| \ge 2\}$ be the set of multiple vertices. When A is independent, any vertex of nonempty type is outside $A \cup \{r\}$. Let B consist of the vertices x incident with an edge $ax \in K$; these are the selected tips.

Lemma 3.1 (Root identities). Assume A is independent. For $xy \in O$,

$$R \subseteq T(x) \cup T(y). \tag{3.3}$$

Moreover,

$$d_O(x) + |T(x)| \le 4, \qquad \sum_{x \in P} |T(x)| \le 3|A| \le 12, \qquad |P| \le 6. \tag{3.4}$$

Write $L_a = \sum_{x \ne r,\ a \in T(x)} (|T(x)| - 1)$ for the load at a selected label. The original development proves four specialized estimates. With $c(Q) \ge j$ meaning $c(e) \ge j$ for every $e \in Q$, these are

Hypothesis	**Specialized load bound (3.5)**
$c(Q) \ge 23$	$L_a \le 1$
$c(Q) \ge 22$	$L_a \le 2$
$c(Q) \ge 21$	$L_a \le 3$
$c(Q) \ge 20$, with A independent	$L_a \le 4$

The last implication requires independence of A. For the first three, it is sufficient that the selected root edge has the stated threshold, even without independence of the entire root neighbourhood.

Proof. An outer edge xy conflicts with every selected ra. Since its endpoints are outside $A \cup \{r\}$, this requires xa or ya. Thus $R \subseteq T(x) \cup T(y)$. The neighbours counted by $d_O(x)$ are outside A, disjoint from those counted by $T(x)$, proving $d_O(x) + |T(x)| \le 4$. Each label has at most three neighbours other than r; counting label--multiple-vertex incidences gives $\sum_{x \in P} |T(x)| \le 12$ and $|P| \le 6$.

To prove the specialized bounds, fix the selected edge ra. Let $X(ra)$ consist of ordered pairs (b, x) with $b \in N_G(r) \setminus \{a\}$, $x \in N_G(a) \setminus \{r\}$ and $bx \in E(G)$. These are the ordered opposite edges of four-cycles through ra.

At thresholds 23, 22 and 21, the cover has exactly six vertices: a smaller cover would give $c(ra) \le 20$. Consequently both endpoints have degree four and their two erased neighbour sets are disjoint. The map $(b, x) \mapsto bx$ is injective on $X(ra)$, since swapping the endpoints would violate this disjointness. Each image is an original edge inside $C(ra)$. Inequality (2.1) bounds the number of internal edges by one at threshold 23, two at threshold 22, and three at threshold 21. Thus $|X(ra)| \le 1, 2, 3$, respectively.

At threshold 23, the implementation also uses the following explicit uniqueness statement: any two ordered opposite edges are equal. Indeed their unordered images lie in a set of size at most one, and disjointness fixes the orientation. At thresholds 22 and 21, it uses the corresponding cardinality bounds on the full ordered family.

For threshold 20, a triangle is still possible and the six-vertex argument cannot be used. Instead assume A independent. Then $A \setminus \{a\}$ and $N_G(a) \setminus \{r\}$ are disjoint, so the same injection is valid. Inequality (2.1), using only $|C(ra)| \le 6$, bounds the internal edges by four. Finally, for each fixed x adjacent to a other than r, the possible first coordinates are exactly $T(x) \setminus \{a\}$. Hence $|X(ra)| = L_a$, proving all four specialized assertions. ■

All these families use original neighbours, including vertices outside the support of Q. The first bound is implemented through ordered-pair uniqueness; the other thresholds retain separate cardinality and root-load declarations. No threshold-parametric theorem is assumed in this route.

Lemma 3.2 (Independence around a high-degree root). Suppose every selected edge has conflict degree at least 21 and $s \ge 3$. Then $|A| = 4$ and A is independent.

Proof. A selected edge incident with r gives $d_G(r) = 4$ by Lemma 2.2. Every pair of vertices of the four-element set A has at least one endpoint in R, since $|R| \ge 3$. An edge inside A would therefore form a triangle with a selected edge incident with r, contradicting Lemma 2.2. ■

Lemma 3.3 (A singleton controls the selected tips). Assume A is independent. If $ax \in K$ and $T(x) = \{a\}$, then every $y \in B$ with $a \notin T(y)$ is adjacent to x. Consequently $|B| \le 6$.

Proof. Choose $by \in K$. As $a \notin T(y)$, we have $b \ne a$. The edges ax and by have no common endpoint. Independence of A, the singleton type of x, and the exclusion of a from $T(y)$ rule out, respectively, the cross-edges ab, xb and ay. Their required conflict must therefore be witnessed by xy. Thus

$$B \subseteq (N_G(a) \setminus \{r\}) \cup (N_G(x) \setminus \{a\}),$$

whose two sets each have size at most three. ■

Lemma 3.4 (Fully selected doubles). Assume A is independent. Call a double x fully selected if both edges from x to $T(x)$ belong to K, and let D_2 be the set of such vertices. Then $|D_2| \le 4$.

Proof. If two fully selected doubles x, y have different types, choose $a \in T(x) \setminus T(y)$ and $b \in T(y) \setminus T(x)$. Conflict between ax and by forces xy, since all other possible witnesses are excluded by the two type differences and independence of A.

If all vertices of D_2 have the same type, choose a label in that type. Its at most three neighbours other than r contain D_2, so $|D_2| \le 3$. Otherwise choose $x, y \in D_2$ with different types. Every $z \in D_2$ has a type different from at least one of $T(x), T(y)$, so is adjacent to at least one of x, y. This includes $z = x$ or $z = y$, since xy is an edge. All such vertices lie outside A. Each of x, y has at most two neighbours outside A, yielding $|D_2| \le 4$. ■

Lemma 3.5 (Two outer-edge bounds). Suppose $s = |A| = 4$ and A is independent. If every type has size at most two, then $|O| \le 4$. If there is a vertex $x_0 \ne r$ such that every type other than $T(x_0)$ has size at most two, then at most four outer edges avoid x_0.

Proof. Let O' be all of O in the first case and the outer edges avoiding x_0 in the second case. In either case all endpoints of its edges have type size at most two. Equation (3.3) implies that the endpoint types of each edge in O' are disjoint two-element sets whose union is A. Both endpoints lie in P. Every vertex of P has at least two neighbours in A, so it has at most two neighbours in P, because $A \cap P = \varnothing$. Fix $a \in A$ and put $D = \{x \in P : a \in T(x)\}$. Each edge of O' has exactly one endpoint in D. Lemma 2.4, applied with the full original graph on P, proves the bound. ■

Lemma 3.6 (All types at most two). Suppose $s = |A| = 4$, A is independent, and $|T(x)| \le 2$ for all $x \ne r$. Then $q \le 18$.

Proof. If there is a selected singleton tip, Lemma 3.3 gives $|B| \le 6$. Otherwise $B \subseteq P$, so (3.4) gives the same bound. Every selected tip supports one edge of K, with a second edge possible only for a fully selected double. Therefore $k \le |B| + |D_2| \le 10$ by Lemma 3.4. Lemma 3.5 gives $o \le 4$. The covering inequality (3.1) yields $q \le 4 + 10 + 4 = 18$. ■

4. The five structural bounds

We now prove the estimates used in the packing argument. All degrees in the hypotheses of the following theorem are full degrees in H_G, irrespective of the selected clique.

Theorem 4.1 (Clique bounds by full conflict degree). Let $\Delta(G) \le 4$ and let Q be a conflict clique. For each row of the following table, the condition $c(e) \ge t$ for every $e \in Q$ implies $|Q| \le b_t$.

Threshold t	**20**	**21**	**22**	**23**	**24**
Bound b_t	19	18	17	16	13

The cases are proved in the next five subsections. They use no regularity, girth, or nonemptiness hypothesis on the original graph.

4.1. Conflict degree at least 24

Proposition 4.2. If every $e \in Q$ has $c(e) \ge 24$, then $q \le 13$.

Proof. Inequality (2.1) shows that a selected edge has a six-vertex cover with no internal edge. Such an edge belongs to no triangle and no four-cycle: a triangle reduces the cover size, and the edge opposite it in a four-cycle lies inside its cover.

Fix a vertex u and consider selected edges ax with $a \in N_G(u)$ and $x \ne u$. We call all such selected edges with a fixed a a branch at u. Each branch has at most three edges. We claim that a branch containing two edges excludes any nonempty branch with a different label.

Suppose ax_1, ax_2, by are selected, where a, b are distinct neighbours of u and $x_1 \ne x_2$. For either i, the equality $x_i = b$ or $y = a$ would put a selected edge in the triangle uab; equality $x_i = y$ would put ax_i in a four-cycle through u. Thus these selected edges have disjoint endpoints. If ab were an edge, both u and b would belong to $C(ax_i)$, with the original edge ub internal to this cover, which is impossible. The cross-edge ay would form a four-cycle $u - a - y - b - u$ containing by, and bx_i would form a four-cycle containing ax_i. All are excluded. Their conflict must be witnessed by x_iy. But then $a - x_1 - y - x_2 - a$ is a four-cycle containing a selected edge, again impossible.

Hence the union of branches with any prescribed set of at most three labels has at most three edges: either all nonempty branches are singletons, or only one branch is nonempty and it has size at most three.

If Q is nonempty, choose $uv \in Q$. Besides uv, there are at most three selected edges incident with u and at most three incident with v. Every remaining selected edge meets $C(uv)$ by Lemma 2.1. Those meeting $N_G(u) \setminus \{v\}$ belong to at most three branches at u and number at most three; the analogous family at v also numbers at most three. Counting these covering families, with any overlaps harmless, gives $q \le 1 + 3 + 3 + 3 + 3 = 13$. ■

4.2. Conflict degree at least 23

Proposition 4.3. If every $e \in Q$ has $c(e) \geq 23$, then $q \leq 16$.

Proof. If all selected degrees are at most two, Lemma 2.3 gives $q \leq 13$. Otherwise take a vertex r of maximum selected degree, so $s = 3$ or $s = 4$. Lemma 3.2 gives an independent four-element neighbourhood A. The corresponding specialized bound in (3.5) gives $L_a \leq 1$ for each $a \in R$.

No type has size at least three. Indeed, such a type meets R, and at a common label it contributes at least two to the load. Also, two distinct multiple vertices cannot have a common label in R, since they would contribute at least two to its load.

We first show $o \leq 2$. If there is an outer edge xy, (3.3) and $|R| \geq 3$ imply that one endpoint, say x, contains at least two labels a, b of R in its type. Consider an outer edge avoiding x. To cover a and b, its endpoints must include vertices having these labels in their types. Any such vertex other than x must be a singleton, by the preceding load observation. Thus the two endpoints would have types $\{a\}$ and $\{b\}$, leaving a third label of R uncovered. This is impossible. All outer edges therefore meet x, and (3.4) bounds their number by two.

If $s = 3$, maximality of the selected degree gives $\sum_{a \in A} d_Q(a) \leq 12$. Equation (3.2) then gives $q \leq 14$.

If $s = 4$, each label in $A = R$ occurs in the type of at most one multiple vertex. The total number of selected edges from A to multiple vertices is therefore at most four. If singleton tips occur, their number is at most six by Lemma 3.3; otherwise their contribution is zero. Hence $k \leq 6 + 4 = 10$, and $q \leq 4 + 10 + 2 = 16$. ■

4.3. Conflict degree at least 22

Proposition 4.4. If every $e \in Q$ has $c(e) \geq 22$, then $q \leq 17$.

Proof. The case of maximum selected degree at most two follows from Lemma 2.3. Otherwise choose a root of maximum selected degree $s \in \{3, 4\}$. Its neighbourhood is independent and has four vertices. For every selected root label, (3.5) gives $L_a \leq 2$.

Case 1: $s = 4$. If some vertex x has $|T(x)| \geq 3$, its type has exactly three elements, since a type of size four exceeds the load bound at each of its labels. Any other multiple vertex has a type meeting $T(x)$; at a common label their load is at least $2 + 1 = 3$, impossible. Thus x is the only multiple vertex. Every outer edge covers four labels and must meet a multiple vertex, hence must meet x. It follows that $o \leq 1$. Since at most sixteen selected edges meet A, (3.2) gives $q \leq 17$.

We may therefore assume all types have size at most two. Every label occurs in at most two multiple types. The sum of the type sizes is at most eight, giving $|P| \leq 4$. Both endpoints of every outer edge belong to P, and each has outer degree at most two. The handshake count gives $2o \leq 2|P| \leq 8$, hence $o \leq 4$.

If there is no selected singleton tip, then $k \leq 2|P| \leq 8$, giving $q \leq 16$. Otherwise choose a selected singleton ax. Lemma 3.3 gives $|B| \leq 6$, and each multiple selected tip contributes at most one extra edge beyond its first. Therefore $k \leq |B| + |P| \leq 10$, so $q \leq 18$.

Suppose equality $q = 18$ holds. The preceding bounds force $k = 10$, $o = 4$ and $|P| = 4$. Every vertex of P then has outer degree exactly two. Together with its two neighbours in A, these exhaust all four possible neighbours. In particular no vertex in P is adjacent to the singleton x, since the two outer neighbours already belong to P. Lemma 3.3 therefore implies that every multiple selected tip has a in its type. At most two multiple vertices contain a, by $L_a \leq 2$. There can be at most two extra selected edges beyond the first edge at each tip, so $k \leq |B| + 2 \leq 8$, contradicting $k = 10$. Thus $q \leq 17$.

Case 2: $s = 3$. Call a vertex heavy if its type contains at least two labels of R, and write Z for the set of heavy vertices. A heavy vertex contributes at least one unit to L_a at each of at least two labels $a \in R$. Since $\sum_{a \in R} L_a \le 6$, there are at most three heavy vertices. Every outer edge covers the three labels of R, so has a heavy endpoint. Each heavy vertex has outer degree at most two. Hence $o \le 2|Z| \le 6$. The selected degree bound three gives $\sum_{a \in A} d_Q(a) \le 12$, and consequently $q \le 18$.

Suppose again that $q = 18$. Equality forces $o = 6$, $|Z| = 3$, and outer degree two at each heavy vertex. Therefore every heavy vertex has type size exactly two, both labels lying in R. Its contributions to the three loads total two. The three heavy vertices saturate all three load bounds, so every label of R occurs in exactly two heavy types.

Every multiple type meets R, since $|A \setminus R| = 1$. A multiple vertex outside Z would add positive load at a label already saturated by Z. Thus all multiple vertices are heavy. Each heavy vertex has two selected outer edges; because every selected degree is at most three, it supports at most one edge of K. A singleton tip also supports at most one edge of K. If singleton tips occur, Lemma 3.3 now gives $k \le |B| \le 6$. If none occurs, $B \subseteq Z$ gives $k \le 3$. In either event $q \le s + k + o \le 3 + 6 + 6 = 15$, contrary to $q = 18$. This excludes equality and proves $q \le 17$. ■

4.4. Conflict degree at least 21

Proposition 4.5. If every $e \in Q$ has $c(e) \ge 21$, then $q \le 18$.

Proof. If every selected degree is at most three, Lemma 2.3 applies. Otherwise choose r with $s = 4$. Its four-element neighbourhood is independent, and (3.5) gives $L_a \le 3$ for all $a \in A$.

Suppose first that every type has size at most two. At a selected label a, every multiple vertex containing a contributes at least one to L_a. The threshold-twenty-one load bound therefore gives at most three such multiple vertices. Every label is selected because $s = 4$. Double-counting incidences gives

$$2|P| \le \sum_{x \in P} |T(x)| = \sum_{a \in A} |\{x \in P : a \in T(x)\}| \le 3|A| \le 12.$$

Thus $|P| \le 6$. If there is a selected singleton tip, Lemma 3.3 gives $|B| \le 6$; otherwise $B \subseteq P$ gives the same bound. Lemma 3.4 bounds fully selected doubles by four, so $k \le |B| + |D_2| \le 10$. Lemma 3.5 gives $o \le 4$. The covering inequality (3.1) yields $q \le 4 + 10 + 4 = 18$. This is the specialized high21 small-type argument: its multiple-vertex bound is obtained from the high21 loads. The threshold-free count in Lemma 3.6 is used later in the high20 case.

Otherwise choose x with $|T(x)| \ge 3$. There is at most one such vertex, because two types of size at least three intersect, contributing at least four at a shared label. All other multiple types have size two. At a label of $T(x)$, there is at most one multiple vertex other than x: two would contribute at least $2 + 1 + 1 = 4$.

An outer edge incident with x is one of at most one such edges, by (3.4). If there is an outer edge yz avoiding x, its endpoint types are complementary doubles by (3.3). Every multiple vertex $w \ne x$ has a type meeting $T(x)$. A label in that intersection belongs to exactly one of $T(y), T(z)$. The preceding uniqueness at that label forces $w = y$ or $w = z$. Thus y, z are the only possible multiple vertices other than x. Every outer edge avoiding x has two multiple endpoints and consequently must be the single edge yz. There are at most two outer edges in total. At most sixteen selected edges meet A, so $q \le 16 + 2 = 18$. ■

4.5. Conflict degree at least 20

Proposition 4.6. If every $e \in Q$ has $c(e) \geq 20$, then $q \leq 19$.

Proof. If all selected degrees are at most three, Lemma 2.3 gives the stronger bound eighteen. Otherwise choose a root r with $s = 4$, so $A = R$ has four vertices. We may assume $q \geq 20$ and derive a contradiction.

We first prove that A is independent; the threshold twenty alone does not exclude triangles. Suppose ab is an edge inside A. The selected edge ra lies in a triangle, so Lemma 2.2 says that $C(ra)$ has no internal edge. If $x \neq r$ and $a \in T(x)$, any $c \in T(x) \setminus \{a\}$ would give an internal edge cx of $C(ra)$. Hence $T(x) = \{a\}$. Interchanging a, b gives the corresponding assertion for b. An outer edge must cover all four labels of R. To cover a, b, its endpoints would have to have the singleton types $\{a\}$ and $\{b\}$, which cannot cover four labels. Thus $o = 0$, and (3.2) gives $q \leq 16$, a contradiction. This proves independence.

The threshold-twenty clause of (3.5) now supplies the load bound $L_a \leq 4$ for every $a \in A$. We distinguish all possible sizes of the types.

Case 1: a type has size four. Choose x with $T(x) = A$. Any other type of size at least three would contribute, together with x, at least $3 + 2 = 5$ at a shared label. Thus every other multiple vertex is a double. At each label there is at most one such other multiple vertex, since x already contributes three units.

No outer edge meets x, as all its neighbours lie in A. If an outer edge yz exists, its endpoint types are complementary doubles. Any other multiple vertex has a label in common with one of these types, and uniqueness at that label forces it to be y or z. Therefore there is at most one outer edge avoiding x. Partition O into edges incident with x and edges avoiding x. The degree inequality $d_O(x) + |T(x)| \leq 4$ bounds the former family (in fact, it is empty). The original branch interface records the sufficient estimate $o \leq 2$, and (3.2) gives its declared bound $q \leq 18$, a contradiction. The stronger consequence seventeen is not needed by this interface.

Case 2: there are two triple types and no type of size four. Let x, y be two distinct triples. Their types share at least two labels; choose distinct shared labels a, b. At either shared label, x and y already contribute $2 + 2 = 4$. Hence any other vertex containing that label in its type is a singleton.

An outer edge avoiding x, y would have to cover a, b using two singleton endpoints of types $\{a\}, \{b\}$. This fails to cover the other two labels. Therefore every outer edge meets x or y. Each has at most one neighbour outside A, so $o \leq 2$. Again $q \leq 16 + 2 = 18$, a contradiction.

Case 3: exactly one triple type and no type of size four. Let x be the unique triple, and let a_0 be the label in $A \setminus T(x)$. Every outer edge avoiding x has complementary double endpoint types. Lemma 3.5 bounds these edges by four. At most one outer edge meets x, giving $o \leq 5$.

We claim $k \leq 10$. The total number of original edges from A to vertices other than r is at most twelve. If at most one of the three edges between x and $T(x)$ is selected, at least two of these original edges are absent from K, proving $k \leq 10$.

Otherwise choose two selected edges ax, bx with distinct $a, b \in T(x)$. Consider any selected edge $a_0y \in K$. We have $y \neq x$, and the type of y has size at most two. As it contains a_0, it misses at least one of a, b, say a. The selected edges ax and a_0y have no common endpoint. Their possible cross-edges aa_0, xa_0 and ay are excluded by independence of A and by the types. Their conflict forces xy. But x has at most one neighbour outside A, so there is at most one possible selected edge $a_0y \in K$. Each of the other three labels supports at most three edges of K. Therefore $k \leq 1 + 3 + 3 + 3 = 10$, as claimed.

Now the covering inequality (3.1) gives $q \leq 4 + 10 + 5 = 19$, contradicting $q \geq 20$.

Case 4: every type has size at most two. Lemma 3.6 gives $q \le 18$, the final contradiction. These four cases exhaust all type configurations, completing the proof. ■

Propositions 4.2-4.6 prove every row of Theorem 4.1. In particular, the last row gives thirteen, although the final counting argument only needs fifteen.

5. Private neighbourhoods and the packing argument

We temporarily work with an arbitrary finite simple graph J. Closed neighbourhoods are denoted by $N_J[v]$. A maximum independent set means one of largest cardinality, not merely an inclusion-maximal set.

Lemma 5.1 (Optimal independent set and private vertices). Choose a maximum independent set I in J for which $\sum_{v \in I} d_J(v)$ is as small as possible. For $v \in I$ define

$$P_v = \{x \in V(J) : N_J[x] \cap I = \{v\}\}.$$

Then $v \in P_v \subseteq N_J[v]$, the set P_v is a clique, and

$$d_J(x) \ge d_J(v) \quad \text{for every } x \in P_v. \tag{5.1}$$

Proof. A choice with both optimization properties exists because there are finitely many subsets. Since I is independent, v is private to itself. The defining intersection implies $x = v$ or x is adjacent to v, proving the containment.

If distinct $x, y \in P_v$ were nonadjacent, neither could equal v. Replacing v by x, y would give an independent set larger than I, because each has no neighbour in $I \setminus \{v\}$. Thus P_v is a clique.

For $x \in P_v \setminus \{v\}$, the set $(I \setminus \{v\}) \cup \{x\}$ is independent and has the same size as I. Minimality of the degree sum then implies $d_J(v) \le d_J(x)$. The case $x = v$ is immediate. ■

Lemma 5.2 (Private-neighbourhood count). For a maximum independent set I and the associated private sets,

$$2|V(J)| \le \sum_{v \in I} (d_J(v) + 1 + |P_v|). \tag{5.2}$$

Proof. Put $t(x) = |N_J[x] \cap I|$. Maximality of I gives $t(x) \ge 1$ for every vertex. The sets P_v are pairwise disjoint and their union consists exactly of the vertices with $t(x) = 1$. A vertex in that union contributes one to $t(x)$ and one to the private-set count; every other vertex contributes at least two to $t(x)$. Thus

$$2|V(J)| \le \sum_{x \in V(J)} t(x) + \sum_{v \in I} |P_v|.$$

Double-counting pairs (x, v) with $v \in N_J[x] \cap I$ gives $\sum_x t(x) = \sum_{v \in I} (d_J(v) + 1)$, which proves (5.2). ■

Lemma 5.3 (A general twenty-packing criterion). Suppose $\Delta(J) \le 24$. Assume that every clique whose vertices all have degree at least t has size at most b_t, where

$$b_{20} = 19, \quad b_{21} = 18, \quad b_{22} = 17, \quad b_{23} = 16, \quad b_{24} = 15.$$

Then $\alpha(J) \ge |V(J)|/20$.

Proof. Take I as in Lemma 5.1. For $v \in I$ write $d = d_J(v)$. If $d \le 19$, the containment $P_v \subseteq N_J[v]$ gives

$$d + 1 + |P_v| \le 2(d + 1) \le 40.$$

If $20 \le d \le 24$, the private set is a clique all of whose vertices have degree at least d, by (5.1). Hence $|P_v| \le b_d = 39 - d$, and again $d + 1 + |P_v| \le 40$. Substituting into (5.2) yields $2|V(J)| \le 40|I|$. Since $|I| = \alpha(J)$, this is the assertion. ■

Proof of Theorem 1.1. Form H_G using the full original graph and let $J = H_G[F]$. Lemma 2.1 gives $\Delta(H_G) \le 24$, and therefore $\Delta(J) \le 24$.

Let Q be a clique in J whose vertices have degree at least t in J, where $20 \le t \le 24$. It is also a clique in H_G, and for each $e \in Q$ we have $c(e) = d_{H_G}(e) \ge d_J(e) \ge t$. Theorem 4.1 applies. Its five bounds meet those in Lemma 5.3, with the stronger last bound $13 \le 15$. Thus J has an independent set M with $|F| = |V(J)| \le 20|M|$. Its vertices are edges from F, and their independence in J says that they are pairwise nonconflicting in the original graph. Therefore M is the required induced matching in G. If F is empty, the same argument or the choice $M = \varnothing$ applies. Sharpness is proved next. ■

6. Sharpness and the precise scope of the conclusion

Proposition 6.1 (Sharpness). There is a 4-regular graph B with twenty edges for which every induced matching has size at most one.

Proof. Start with a cycle of length five. Replace each vertex by an independent pair V_i, indexed modulo five, and join every vertex of V_i to every vertex of V_{i+1}. There are ten vertices, each with four neighbours, and twenty edges.

Every edge belongs to one of the five complete bipartite graphs between consecutive pairs. Consider two distinct edges. If they belong to the same such graph, they either meet or have cross-adjacent endpoints. If their corresponding cycle edges meet, an endpoint in their common cluster is adjacent to an endpoint of the other edge in a neighbouring cluster. If the corresponding cycle edges are disjoint, some endpoint cluster of one is adjacent on the five-cycle to an endpoint cluster of the other, again giving cross-adjacent endpoints. Thus every pair conflicts. A single edge is an induced matching, so $\nu_s(B) = 1$. Taking $F = E(B)$ in an inequality $|F| \le C|M|$ forces $C \ge 20$. ■

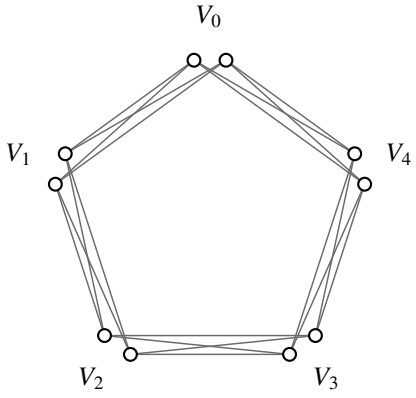


Figure 1. The doubled five-cycle: five independent pairs, ten vertices and twenty edges. Each pair of consecutive clusters induces a complete bipartite graph.

This is the doubled five-cycle already used by Joos and Nguyen [1]; neither the example nor the sharpness of the full-edge-set constant is new here.

The proof establishes a cardinality guarantee separately for each candidate set. It does not establish the analogous inequality for every assignment of arbitrary nonnegative edge weights. In conflict-graph language, the conclusion is the Hall-ratio bound (1.2). The general inequalities $\rho(J) \le \chi_f(J) \le \chi(J)$ point in the opposite direction from a proposed deduction of a fractional or ordinary colouring bound from a

Hall-ratio bound. Known separations between Hall ratio and fractional chromatic number [4] reinforce the need to keep these statements distinct.

The optimization used in Lemma 5.1 is an existence argument. No running-time claim about the accompanying Lean theorem is made. The mathematical proof is independent of a particular implementation of maximum-independent-set search.

Appendix A. Original Lean route and dependency structure

The formal endpoint for this manuscript is **StrongEdge.induced_matching_packing20**. Its original source and configuration files are delivered unchanged. The human argument above expands the finite-set injections, equality cases and exchange arguments into ordinary mathematical prose. It preserves the original proof's one-sided cover interface, ordered four-cycle families, specialized threshold bounds and the deliberately sufficient bound eighteen in the four-element-type high20 branch.

The code's selected root-edge family excludes edges incident with the root, despite the informal phrase “root edge” sometimes meaning an edge incident with r. Here the star consists of selected ra edges and K is the implementation's **rootEdges** family. **rootTips** corresponds to B, **multipleTips** to P, and **outerEdges** to O.

Manuscript result	Original StrongEdge module or declaration
Lemma 2.1, one-sided cover and incidence identity	Bounds/ConflictDegree; Bounds/EdgeCounting
Lemmas 2.3-2.4	Bounds/SubcubicEighteen; Bounds/DegreeTwoClique
Ordered four-cycle uniqueness, threshold 23	Bounds/High23Local; Bounds/High23Support
Specialized loads 22, 21 and 20	Bounds/High22Support; Bounds/High21Support; Bounds/High20Local
Root covering counts, singleton tips	Bounds/RootCounting; Bounds/High22Subcubic
Doubles and complementary outer endpoints	Bounds/High21Double; Bounds/High21RootFour
Proposition 4.2	Bounds/High24: high24_clique_card_le_thirteen
Proposition 4.3	Bounds/High23: high23_clique_card_le_sixteen
Proposition 4.4	Bounds/High22: high22_clique_card_le_seventeen
Proposition 4.5	Bounds/High21: high21_clique_card_le_eighteen
Proposition 4.6	Bounds/High20: high20_clique_card_le_nineteen
Optimality and double count	Basic; PrivateExchange; DoubleCount; Packing
Five-field criterion and candidate restriction	MinimalPacking
Final entry point	High20Packing: induced_matching_packing20

The implementation first proves the high24 bound thirteen and then exposes the weaker fifteen required by **MinimalCliqueBounds20**. The high23, high22, high21 and high20 packing modules successively assemble the remaining fields. The final theorem has no unresolved structural premise. A historical record demanding seven at threshold twenty-four is not used by this endpoint; importing a file containing that record does not mean the proof assumes it.

The private predicate in the code states that $x = v$ or xv is an edge, and that no other member of I equals or is adjacent to x. For $v \in I$ this is exactly $N_J[x] \cap I = \{v\}$. The degree-sum minimization is over maximum independent sets; it does not impose regularity or a minimum degree on G.

The sharpness witness in Section 6 is supplied in a separate checked audit file, because the original final theorem states existence and does not itself encode optimality of the constant. Its additional real-constant theorem does not modify the original sources. The graph is encoded on **Fin 10**, with adjacency between the two-vertex clusters consecutive modulo five.

The delivery pins Lean **4.35.0-rc4** and mathlib **49fd9dee36e7bbf345fe7981d2de6aeeb591f75b**. Original-source hashes are compared with the independent audit snapshot. The final theorem's actual type-and-proof dependency closure is checked for admitted terms, extra axioms, unsafe declarations and unexpected project endpoints. The companion route does not contribute any declaration to this original proof.

The global mathematical strategy is shared with Route A. The main differences are the intermediate interfaces, the organization of local estimates, the reuse of a threshold-free lemma, and the strength of one branch bound. These facts do not justify a claim of two unrelated discoveries. The literature qualifications in Section 1 apply equally to both routes.

References

[1] F. Joos and V. H. Nguyen, *Induced Matchings in Graphs of Degree at Most 4*, SIAM Journal on Discrete Mathematics 30 (2016), no. 1, 154-165. DOI: 10.1137/140986980.

[2] M. Huang, M. Santana and G. Yu, *Strong Chromatic Index of Graphs With Maximum Degree Four*, Electronic Journal of Combinatorics 25 (2018), no. 3, Paper P3.31. DOI: 10.37236/7016.

[3] M. Fürst, M. Leichter and D. Rautenbach, *Locally Searching for Large Induced Matchings*, Theoretical Computer Science 720 (2018), 64-72. DOI: 10.1016/j.tcs.2018.02.006.

[4] R. Steiner, *Fractional Chromatic Number Vs. Hall Ratio*, Combinatorica 45 (2025), Article 37. DOI: 10.1007/s00493-025-00164-0.